FRUCTUAL – An AI-Driven Personalized Fruit-Sugar Formulation and Dispensing System for Fitness and Diabetic Health Management

The field of the invention is healthcare technology and IoT-based smart nutritional management systems, particularly within the domain of the Internet of Medical Things (IoMT).

Abstract

Managing sugar intake is a significant challenge for individuals with diabetes and those pursuing fitness and overall wellness. Conventional sugars and artificial sweeteners often lack adaptability, either posing health risks or failing to meet individual nutritional requirements. This system introduces a smart, personalized approach to sweetness by combining artificial intelligence, nutrition science, and automated dispensing. Natural sugars are extracted from the fruits, which have a low glycemic index and high nutritional value. These are then processed into a dried powder and used to create customized sugar formulations. A machine learning model evaluates user-specific parameters, including age, Body Mass Index(BMI), physical activity level, blood glucose readings, and medical history, to recommend an optimal sugar type and precise dosage. For diabetic users, daily blood glucose inputs enable dynamic adjustments. When elevated levels are detected, the system can reduce the recommended quantity, alter the formulation, or advise temporary avoidance, ensuring day-to-day safety while maintaining long-term personalization. An intelligent dispensing machine then delivers the exact recommended dose—typically a few grams—in powder form, ready for use in food and beverages. This ensures diabetic patients maintain safe sugar levels while fitness-focused users get a natural, healthy energy boost. By integrating real-time health monitoring with smart nutrition dispensing, FRUCTUAL supports preventive healthcare and empowers people to make safer, data-driven dietary choices every day. It's a step toward smarter, sustainable, and human-centered wellness technology.

Literature Review

Existing systems focus individually on glucose monitoring, prediction, dietary recommendation, or automated dispensing. However, there is no integrated system that combines real-time physiological data, intelligent decision-making, and adaptive multi-source sugar dispensing using natural alternatives.

S. No	Paper Title	Authors	Year	Description	Limitations
1	Glycemic Index: Physiological Mechanisms Relating to Obesity, Diabetes, and Cardiovascular Disease	Thomas M. S. Wolever	2020	Explains the role of Glycemic Index in controlling blood glucose and dietary planning	Not personalized; no real-time integration
2	Deep Learning for Blood Glucose Prediction Using CGM Data	Zhu et al.	2021	Uses deep learning models to predict glucose levels from CGM readings	Prediction only; no control or dispensing mechanism

S. No	Paper Title	Authors	Year	Description	Limitations
3	Continuous Glucose Monitoring Systems: Current Status and Future Perspectives	Heinemann & Freckmann	2021	Reviews CGM technology and its applications in healthcare	Only monitoring; lacks automated response system
4	Personalized Glycemic Response Prediction Using Machine Learning	Mendes-Soares et al.	2022	Predicts individual glucose responses based on user-specific data	Requires large datasets; not suitable for real-time embedded systems
5	Smart Food Dispensing System with Sensor-Based Control	Zhang et al.	2025	Automated dispensing system using sensors for controlled output	Fixed logic; no health-based adaptive decision-making

Product Survey

Recent systems exist for glucose monitoring, AI-based prediction, and automated dispensing independently. However, no system integrates real-time CGM data with intelligent decision-making and multi-source adaptive sugar dispensing, which is the novelty of our work.

S. No	Product / System Name	Company / Developers	Year	Description	Limitation
1	Levels Health CGM Platform	Levels Health	~2023–Present	Real-time glucose tracking with personalized metabolic insights via app	No automated action or dispensing system
2	NutriSense CGM + Coaching System	NutriSense Inc.	~2022–Present	Combines CGM data with expert guidance for personalized nutrition	Only advisory; no hardware integration
3	January AI Metabolic Health Platform	January Inc.	~2024	AI-based prediction of blood glucose response to meals	Prediction only; no real-time actuation
4	Xiaomi Smart Pet Feeder 2	Xiaomi	~2023	IoT-enabled feeder with portion control and scheduling via app	No health-based personalization or sensing
5	Smart Multi-Grain Dispenser (Touchless IoT Models)	Various (2024 models)	~2024	Multi-container automated dispensing systems for food grains	No intelligence or physiological input integration

Input Parameters & Justification

1. Static Inputs

- Age
- Height
- Medical history

2. Semi-Dynamic Input

BMI (Body Mass Index)

- Calculated using:

$$BMI = \frac{weight}{height^2}$$

- Updated:
  - Monthly OR
  - When weight change is detected

Justification:
BMI does not change daily and is used for **long-term metabolic assessment**

3. Real-Time Input (Refined with Weightage)

Continuous Glucose Monitoring (CGM)

The system processes CGM data using a **multi-layer weighted model**:

Parameter	Description	Weight
Immediate Glucose Value	Latest CGM reading (last 5–10 min)	**50%**
Short-Term Average	Moving average (30–60 minutes)	**30%**
Historical Trend	72-hour rolling average	**20%**

Justification:

- Ensures fast response (immediate value)
- Maintains stability (short-term avg)
- Captures user pattern (3-day trend)

Why This Weighting is Correct

🔹 **50% Immediate Value**

- Most critical for **real-time safety**
- Prevents dangerous spikes

🔹 **30% Short-Term Average**

- Filters noise/fluctuations
- Gives stability

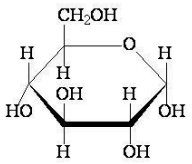
🔹 **20% Historical Trend**

- Captures metabolic behavior
- Prevents repeated poor decisions

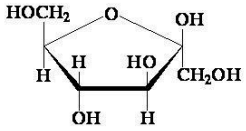
4 .Fruit Sugar Composition Analysis

Fruit sugars are not a single compound but a mixture of:

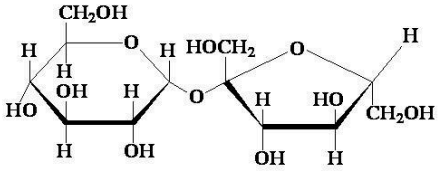
- Glucose ($C_6H_{12}O_6$)
- Fructose ($C_6H_{12}O_6$)
- Sucrose ($C_{12}H_{22}O_{11}$)



glucose



fructose



sucrose

Fruit-derived sugars are utilized in place of refined sugar due to their compositional diversity, lower glycemic impact, and improved metabolic compatibility. Unlike refined sucrose, fruit sugars consist of varying proportions of glucose and fructose, enabling controlled absorption and facilitating personalized sugar delivery based on real-time physiological parameters.

Parameter	Normal Sugar (Refined Sugar)	Fruit Sugar (Natural Sugar)
Chemical Form	Pure **Sucrose ($C_{12}H_{22}O_{11}$)**	Mixture of **Glucose ($C_6H_{12}O_6$), Fructose ($C_6H_{12}O_6$), Sucrose ($C_{12}H_{22}O_{11}$)**
Structure	Fixed composition	Variable composition (depends on fruit)
Glycemic Index (GI)	High (rapid spike)	Low to Medium (controlled release)
Absorption Rate	Fast	Slower (due to fructose & fiber effect)
Blood Glucose Impact	Sudden increase	Gradual increase
Nutritional Value	No nutrients (“empty calories”)	Contains vitamins, antioxidants, micronutrients
Fiber Content	None	May retain traces (depending on processing)

Parameter	Normal Sugar (Refined Sugar)	Fruit Sugar (Natural Sugar)
Metabolic Effect	Direct glucose spike	Balanced metabolic response
Health Impact	Increases risk of diabetes, obesity	Safer alternative when controlled
Customization	Not customizable	Can be selected based on GI & user condition

5. Fruit Sugar Classification

Step 1: Using the Existing WGS (Base Input)

$$WGS = (0.5 \cdot G_{current}) + (0.3 \cdot G_{short}) + (0.2 \cdot G_{3day})$$

👉 This gives overall glucose state

Step 2: Add "Decision Score"

We now combine all factors into a **Sugar Selection Score (SSS)**

$$SSS = W_1 \cdot G_{norm} + W_2 \cdot T_{trend} + W_3 \cdot H_{factor}$$

Where:

1. G_{norm} (Normalized Glucose)

Convert WGS into a normalized value:

$$G_{norm} = \frac{WGS - G_{min}}{G_{max} - G_{min}}$$

👉 Scales between 0 and 1

---

2. T_{trend} (Trend Score)

$$T_{trend} = \frac{G_{3day} - G_{normal}}{G_{normal}}$$

- Positive → consistently high
- Negative → low trend

3. H_{factor} (Health Condition)

Assign values:

Condition	Value
Diabetic	1
Pre-diabetic	0.5
Healthy	0

4.Suggested Weightage

Component	Weight
Glucose (current state)	0.5
Trend (3-day)	0.3
Health condition	0.2

5.Final Formula

$$SSS = (0.5 \cdot G_{norm}) + (0.3 \cdot T_{trend}) + (0.2 \cdot H_{factor})$$

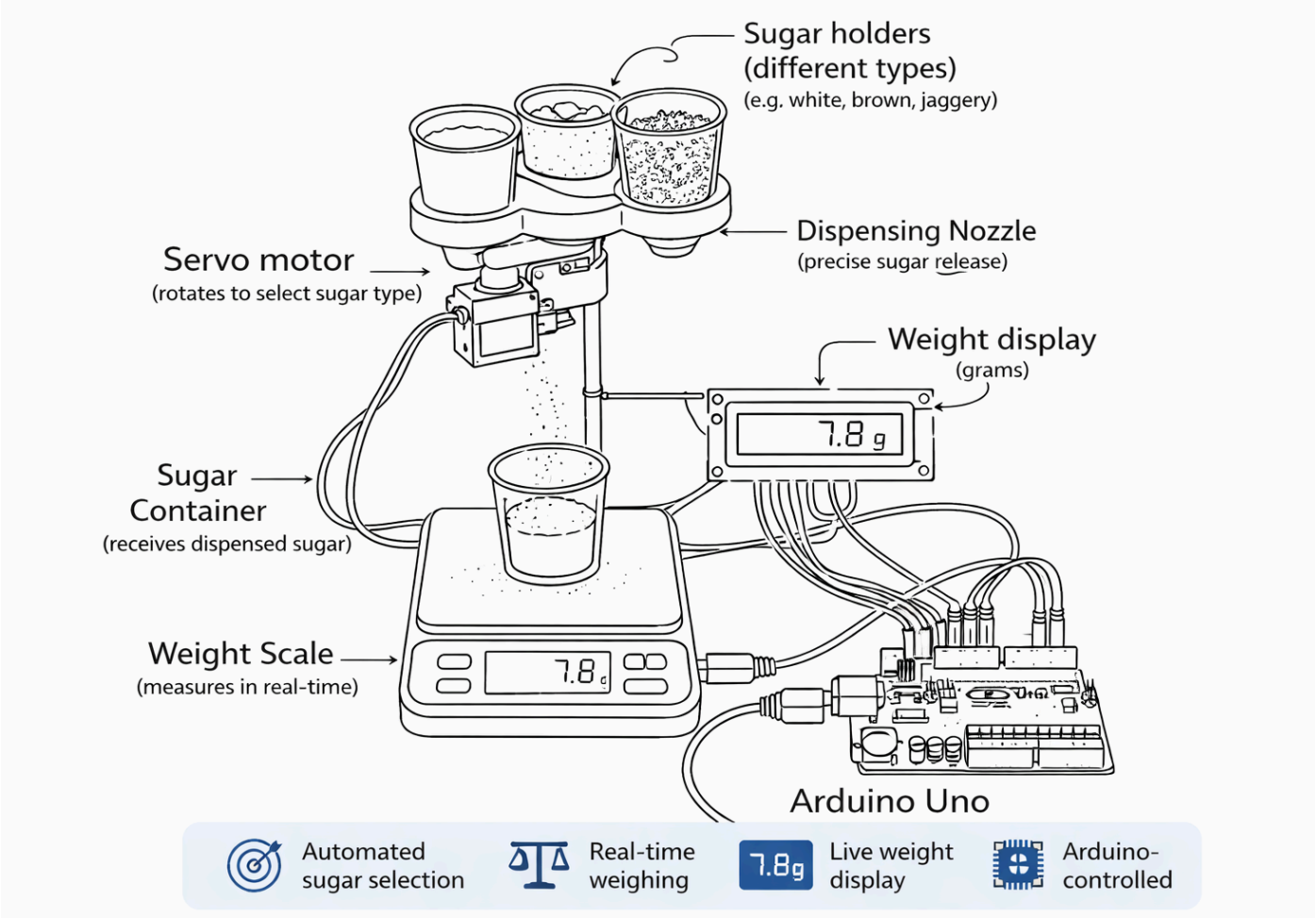
6.Mapping Score → Sugar Type

SSS Value	Sugar Type
≥ 0.7	Low GI (Apple, berries)
0.4 – 0.7	Medium GI (Dates, orange)
≤ 0.4	High GI (Banana, mango)

Simple Interpretation

- Higher score → more restriction → **Low GI**
- Lower score → need energy → **High GI**

Prototype components



S.No	Component Name	Description	Purpose / Function in System
1	Sugar Holders (Different Types)	Containers holding different sugar types (white, brown, jaggery, etc.)	Stores multiple sugar types and allows user selection
2	Rotating Platform / Mount	Base holding all sugar containers	Supports and aligns containers for selection mechanism
3	Servo Motor	Small motor controlled by Arduino	Rotates the platform to select the required sugar type
4	Dispensing Nozzle	Outlet below selected container	Releases sugar in a controlled manner
5	Sugar Container (Cup)	Cup placed below nozzle	Collects dispensed sugar
6	Weight Scale (Load Cell System)	Electronic weighing device	Measures the exact amount of sugar dispensed in real-time
7	Weight Display (LCD)	Digital display screen	Shows weight of sugar (in grams) to the user
8	Arduino Uno	Microcontroller board	Controls entire system (motor, weight reading, display)
9	Connecting Wires	Electrical connections	Transfers signals and power between components
10	Supporting Stand / Frame	Vertical structure holding system	Provides mechanical support and stability

Web Application:

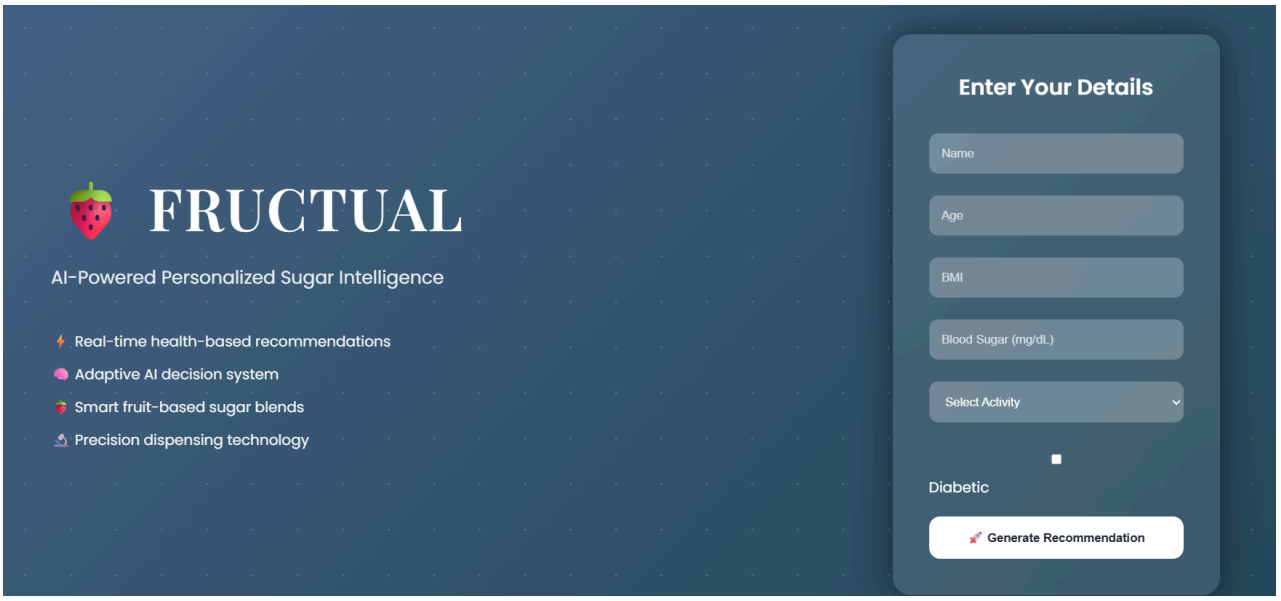


Figure.1

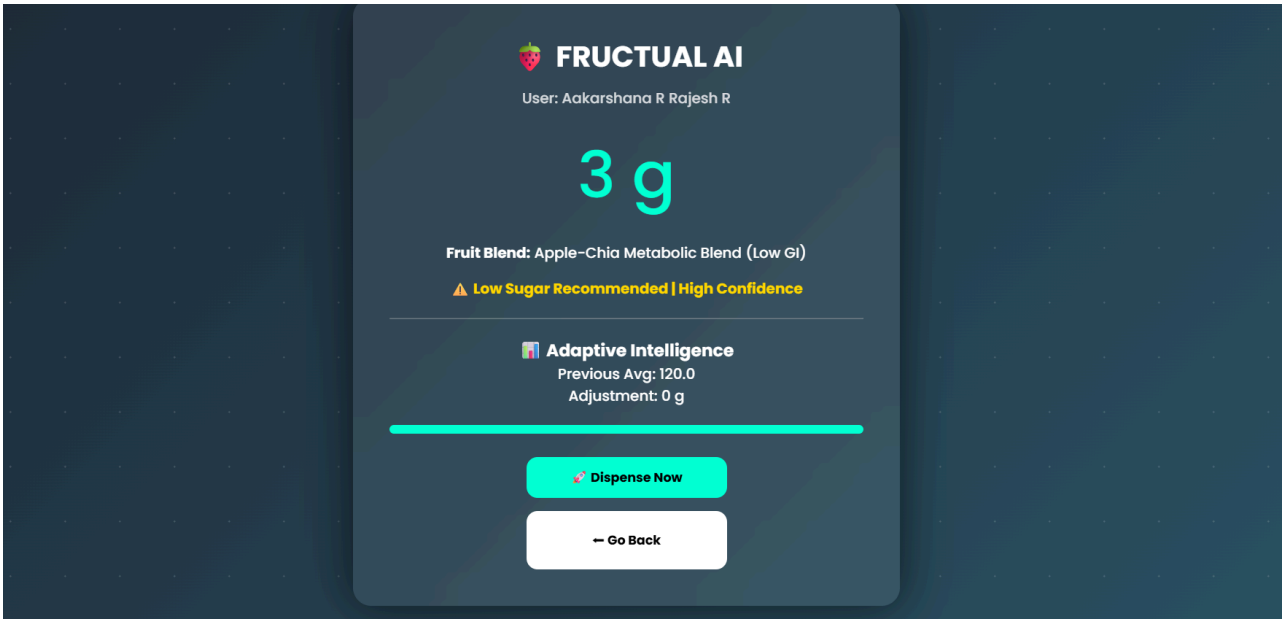


Figure.2

🔹 Figure 1: User Input Interface (Home Screen)

The first screenshot represents the **main user interface of the FRUCTUAL system**, where user-specific health parameters are collected.

Description:

The interface is designed as a **full-screen, visually interactive dashboard** with a split layout:

- **Left Panel**: Displays system branding and key features
- **Right Panel**: Contains the input form

Input Parameters:

The system collects the following user data:

- Name
- Age
- Body Mass Index (BMI)
- Blood Sugar Level (mg/dL)
- Activity Level (Low / Medium / High)
- Diabetic Status (Checkbox)

Purpose:

This interface acts as the **data acquisition layer**, enabling:

- Personalized analysis
- User-specific sugar recommendation
- Adaptive decision-making based on health inputs

Key Features:

- Elegant UI with dynamic background (particle animation)
- Glassmorphism-based input panel for better usability
- Responsive and user-friendly design

🔹 Figure 2: Recommendation & Control Interface

The second screenshot represents the **output interface**, where the system displays the computed recommendation and allows user interaction with the dispensing mechanism.

Description:

After processing the inputs, the system displays:

- **Recommended sugar quantity (in grams)** → prominently shown (e.g., *3 g*)
- **Selected fruit-based sugar blend**
- **Health status message** (e.g., Low Sugar Recommended)
- **Confidence level of the AI decision**
- **Adaptive intelligence insights** (previous average + adjustment)

Interactive Controls:

- **Dispense Now Button**
  → Triggers the hardware (Arduino-controlled dispensing system)
- **Go Back Button**
  → Returns to input interface for new data entry

Additional Feature:

- A **progress indicator bar** representing dispensing readiness / process

Purpose:

This interface serves as:

- Decision visualization layer
- Human-machine interaction point
- Control interface for automated dispensing system

Software Components

🔹 1. Frontend (User Interface Layer)

- Built using: **HTML, CSS**
- Responsibilities:
  - Collect user input
  - Display AI-generated recommendations
  - Provide control buttons (Dispense / Back)
  - Enhance usability through dynamic UI

🔹 2. Backend Server (Application Layer)

- Built using: **Python Flask framework**
- Key functions:
  - Handles HTTP requests (/calculate, /dispense)
  - Processes form data
  - Connects frontend with AI logic
  - Sends commands to hardware (Arduino)

🔹 3. AI Decision Engine (ai_engine.py)

- Core logic module
- Responsibilities:
  - Computes sugar dosage based on:
    - BMI
    - Blood sugar level
    - Activity level

        - Diabetic condition
    - Applies:
        - Rule-based adjustments
        - Historical adaptive learning
    - Selects:
        - Appropriate fruit-based sugar blend
    - Outputs:
        - Recommended grams
        - Status message
        - Confidence level

🔹 4. Database System (SQLite)

- Stores:
    - User details
    - Previous blood sugar logs
    - Recommendation history
- Purpose:
    - Enables **adaptive intelligence**
    - Supports historical data analysis

🔹 5. Hardware Communication Module

- Uses: **PySerial**
- Function:
    - Sends computed sugar quantity to Arduino
    - Controls real-time dispensing mechanism

🔹 6. Embedded System (Arduino)

- Controls:
    - Servo motor (for sugar selection)
    - Dispensing mechanism
    - Load cell (weight measurement)
- Executes:
    - Commands received from software
    - Precise sugar dispensing

System Flow (Simple Explanation)

1. User enters health data → UI
2. Data sent to Flask backend
3. AI engine calculates sugar dosage
4. Result displayed to user
5. User clicks “Dispense”
6. Command sent to Arduino
7. Sugar dispensed accurately

How the Present Invention Overcomes Existing Problems & Its Advantages

How it solves existing problems

Many existing systems either only monitor, only suggest, or only dispense, but they do not provide a complete solution. This invention addresses those issues in a practical and integrated manner:

- Most systems provide general recommendations, whereas this system considers individual factors such as BMI, blood glucose level, physical activity, and medical history to provide personalized sugar intake.
- Existing devices typically only display health data, but this system includes a rule-based mechanism that actively determines the appropriate quantity of sugar.
- In conventional methods, users estimate sugar intake manually, which can lead to overconsumption. This system eliminates that risk by automatically controlling the dispensing process.
- Many dispensers lack precision, but this system uses a load cell to ensure accurate measurement before and during dispensing.
- Instead of relying only on artificial sweeteners, this system supports healthier alternative sugar options.
- Current solutions are often fragmented, but this invention integrates monitoring, decision-making, and dispensing into a single system.

Advantages of the invention

- Personalized operation based on individual health conditions
- High accuracy due to load cell-based measurement
- Fully automated system with minimal user intervention
- Improved safety by preventing excessive sugar intake
- Flexibility to use different types of sugar
- Cost-effective implementation using simple components
- User-friendly interface with clear output display
- Easily extendable for future enhancements such as IoT integration